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Reflection Journal

Brian- Introduction + Understanding of Machine Learning Types

In this lab, I learned about different types of machine learning and their uses. Supervised learning uses data with correct labels, so the model can learn patterns and make predictions. Unsupervised learning works with data that doesn't have labeled answers and instead looks for patterns or groups. Reinforcement learning is different, where an agent learns by doing and receiving either a reward or penalty. I also learned that the machine learning workflow has multiple steps, including preparing the data, selecting features, splitting the data into different training and testing sets, training a model, and evaluating the performance.

One of the main things I learned from the project was how the different parts of the workflow work together. I used the Wine dataset to train the different classification models and then test the models using data that they didn't train on. I learned that the training data is used to learn patterns, while the testing data is used to see how well the model makes predictions on new data. I also learned that choosing the right features can affect how well the model performs. Overall, this lab helped me understand that machine learning is not just about training a model, but also preparing the data, testing the results, and understanding what the model learned.

Toan — Wine Classification Project + ML Workflow

In this wine classification project, I used a chemical analysis dataset of wines grown in three different Italian cultivars to predict their target classes. Among the models I tested, Logistic Regression performed the best because it could cleanly separate the groups using the chemical features.

The machine learning workflow I followed had four main steps. First, for Data, I loaded the dataset, checked for any missing values, and explored the features using distribution plots and correlation matrices. Second, for Training, I split the data and used 80% of it to train the models so they could learn the patterns between the chemical attributes and the wine classes. Third, for Testing, I used the remaining 20% of unseen data to test how well the models could make predictions on new information. Finally, for Evaluation, I measured performance using accuracy scores, classification reports, and confusion matrices to see exactly where the model got things right or wrong.

To improve performance next time, I could try selecting different combinations of features, adjusting the model settings, or testing other algorithms like decision trees with different depth limits. My main takeaway from this project is that getting a high score during training does not mean much if the model fails on new data. Following every single step of the workflow carefully and testing thoroughly on unseen data is essential for building a reliable machine learning model.

Montserrat — Real-World Application + Key Learnings

Machine learning could be used in cybersecurity to detect suspicious network activity and identify possible cyberattacks. A model could learn the difference between normal and malicious activity and then use those patterns to detect potential threats. This would be a supervised

learning problem because the model could learn from labeled examples of normal and malicious activity. The data could include network traffic, login activity, IP addresses, connection information, and examples of both normal and malicious activity. Having labeled examples would help the model learn how to identify suspicious behavior.

The most important concept I learned was the complete machine learning workflow. I learned how data is prepared, how features and targets are selected, how the data is divided into training and testing sets, and how a model is trained and evaluated. The most challenging part for me was understanding the difference between training and testing data and how features and the target are used by a machine learning model. It took some practice to understand how the model learns from the training data and then makes predictions on data it has not seen before. I would like to learn more about how machine learning models improve their accuracy and how feature selection can affect model performance. I also want to explore how machine learning is used with real-world cybersecurity data and how more advanced models, such as neural networks and deep learning, can be applied to larger datasets.

This lab helped me better understand how machine learning works in practice. Before this lab, I understood the basic idea of machine learning, but working with the Wine dataset allowed me to see the process step by step. I learned about supervised, unsupervised, and reinforcement learning and how each type is used for different problems. I also learned how important it is to prepare data, separate training and testing data, train models, and evaluate their performance.